



Theertha Bharathan

Software Engineer

Master's graduate in Information Technology with hands-on experience building and deploying production systems across embedded Linux environments, CI/CD pipelines and backend services. Skilled in Python, Go, Docker and Linux, with a strong focus on test-driven development, automation and building reliable systems that engineering teams depend on daily. Experienced working across distributed systems, APIs and event-driven architectures in agile teams.

Work Experiences

Software Developer (HiWi)

09/2025 – Present

University Medical Center Mainz, Mainz, Germany

- Developed a clinical Android app using Flutter for capturing and processing medical data.
- Designed and deployed a FastAPI based internal tool to dynamically extract, validate, preview, and export clinical research data from SQL Server.

Software Engineer (Work-Study)

11/2023 – 04/2025

Bosch Rexroth, Schwieberdingen, Germany

- Built and deployed Go and Python microservices running in production across 1,000+ embedded Linux field devices.
- Enhanced the CI/CD pipeline using Jenkins for automating package builds and deployments, reducing manual intervention by 60%.
- Applied test-driven development (TDD) enabling robust backend services with automated unit/integration tests and reducing bugs.
- Implemented new features for embedded Linux (Debian-based) devices for improving connectivity and performance.

Intern - Software Developer

04/2023 – 10/2023

Bosch Rexroth, Schwieberdingen, Germany

- Developed automated build toolchains for Python applications, integrating cross-compilation, snap packaging and Docker containerization.
- Evaluated and optimized deployment environments (Docker, Ubuntu, Raspberry Pi) for scalable IoT solutions.

Software Engineer

08/2019 – 10/2022

Mercedes Benz Research & Development India

- Developed an AI-based ECU log analysis system achieving 90% prediction accuracy, accelerating diagnostics and reducing SLA Breaches.
- Designed and tested REST APIs using Flask, enabling seamless machine learning model integration.
- Aligned data requirements between workshop stakeholders, R&D, and support teams to improve data transparency across the service value chain.

Education

Information Technology (Master of Engineering)

10/2022 – 02/2026

Frankfurt University of Applied Sciences, Frankfurt, Germany

Electronics and Communication Engineering (Bachelor of Engineering)

08/2015 – 05/2019

NMAM Institute of Technology, Nitte, India

Projects

- **Master Thesis - Infant Carrier Seat Occupancy Detection using AI:** Python, NumPy, TensorFlow, AI/ML, SHAP, Explainable AI, Hyperparameter tuning, Streamlit - GUI
- **Cloud-Based Anomaly Detection System:** C#, Azure Cloud, Docker, ML
- **Service Management Portal:** Django, PostgreSQL, AWS, Agile/Kanban

Contact

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Skills

Programming & Scripting

- Python, Go, SQL, Bash
- C++ (foundational)
- HTML, CSS, JavaScript

Cloud & DevOps

- Docker, Snapcraft
- Kubernetes
- Jenkins (CI/CD pipeline)
- Linux
- Microsoft Azure, AWS

Backend & API Development

- FastAPI, Flask, Django
- REST API
- Microservices, MQTT
- Event-Driven Architecture

SW Development Practices

- Test-driven development
- Unit Testing & Debugging

Methodologies & Tools

- GIT
- Jira
- Agile/Scrum
- Kanban

Machine Learning & AI

- NumPy, Scikit-learn, Pandas
- TensorFlow, PyTorch
- Explainable AI (SHAP)
- Feature engineering

Awards

Gold Award – Mercedes-Benz R&D India

Recognised by business partners for outstanding project delivery and self-initiated process improvements.

KSCST Research Project

State-funded by the Karnataka State Council of Science & Technology. Selected for the state-level KSCST seminar.

Languages

- English (Expert – C1)
- German (B1/B2 — actively improving)
- Malayalam (Native)